

Saad Rafiq

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EDUCATION

Master of Science in Computer Science
Georgia Institute of Technology

Expected Graduation: May 2028

Bachelor of Science in Computer Science
Bachelor of Science in Applied Mathematics
Texas State University – San Marcos, TX | **GPA:** 3.8

Graduated: December 2024

RELEVANT COURSEWORK

Data Structures & Algorithms - Machine Learning - Object-Oriented Programming - Parallel Programming
Software Engineering - Computer Vision

TECHNICAL SKILLS

Programming Languages - C++, Python, Bash, Java, Groovy, MATLAB

Simulation & Robotics: ROS/ROS2, SLAM, LiDAR, pymavlink, dronekit, Depth Cameras, Nvidia Jetson, Gazebo

Infra & Tooling: Linux, multithreading, subprocess, threading, Git, Docker, Jenkins, AWS, Azure

AI / Perception: OpenCV, PyTorch, TensorFlow

EXPERIENCE

Software Engineer I – Hopkinton, Massachusetts

Dell Technologies

February 2025 – Present

- Developed CI/CD pipelines for the Dell Lightning storage platform, improving build and deployment consistency
- Led packaging and integration for the product's first release, consolidating components from multiple engineering teams into a single, production ready deliverable
- Owned the automation and build process for Nvidia DOCA OFED drivers, ensuring reliable installation and compatibility across target Linux environments
- Automated performance testing using Bash and Python, significantly reducing manual testing effort and freeing DevOps engineers for higher value work

Software Engineering Intern – Austin, Texas

Dell Technologies

May 2024 – August 2024

- Developed a Python SDK for AWS, Azure, and GCP platforms for Dell APEX Protection Storage (DDVE)
- Designed extensive test cases for each platform and REST API functionality of the SDK to ensure reliability
- Applied object-oriented programming principles to design a scalable, maintainable, and modular SDK
- Collaborated closely with architects and managers to fit customer needs and enhance effectiveness in the market

CTO & Co-Founder – San Marcos, Texas

Firefly Robotics, LLC

January 2023 – November 2025

- Built three fully autonomous custom drones powered by Nvidia Jetson Nanos to detect people in burning building scenarios for firefighters, supporting faster firefighter rescue operations
- Developed sensor-driven navigation and perception pipelines, accounting for noise, latency, and degraded visibility
- Implemented SLAM based autonomy enabling drones to navigate unknown, dynamic indoor environments
- Collaborated across electrical, mechanical, and materials teams to ensure simulation matched real-world behavior

Software Engineering Intern – San Marcos, Texas

High-Performance Engineering (HiPE) Lab

June 2022 – January 2023

- Utilized 3 Nvidia Jetson Nanos to sense, acquire, and transmit environmental analytics for a firefighting robot
- Implemented real-time data pipelines using ROS2 pub/sub and action-client architectures
- Integrated AI models for audio perception, processing sensor data via subprocess pipelines
- Designed a custom Ubuntu image that enabled CUDA for machine learning optimization

PUBLICATIONS

A. Rafiq et al., "Design of Autonomous Rover for Firefighter Rescue: Integrating Deep Learning with ROS2," 2024 IEEE World AI IoT Congress (AIIoT), Seattle, WA, USA, 2024, pp. 421-428, doi: 10.1109/AIIoT61789.2024.10578950.

A. Rafiq, et al., "Saving lives while reducing first responder risks – with AI," [White paper] Dell Technologies, 2023.

<https://workstationguides.com/briefs/texasstateuniversity/>